

EXPERIMENT 9

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Experiment 9– Implement Connection using TCP

Objective

- To understand the basic working principles and characteristics of UDP (User Datagram Protocol).
- To implement UDP client-server communication using Python socket programming.
- To establish communication between server.py and client.py using UDP sockets.
- To understand the process of sending and receiving datagrams without establishing a dedicated connection between the client and server.

Theory:

UDP (User Datagram Protocol) is a connectionless transport-layer protocol used for communication between devices over a network. Unlike TCP, UDP does not establish a connection before transmitting data and does not guarantee delivery, ordering, or retransmission of packets. In UDP communication, data is sent as independent **datagrams**, where each datagram contains the destination IP address and port information. In this experiment, Python's socket module is used to implement UDP communication. The server creates a UDP socket and binds it to a specific IP address and port using `bind()`, while the client creates a socket and sends data to the server using `sendto()`. The server receives the data using `recvfrom()` and can send a response back to the client. Since UDP is connectionless, there is no three-way handshake or connection establishment process. This makes UDP faster and suitable for applications such as streaming, online gaming, DNS, and real-time communication where low latency is more important than guaranteed delivery.

Practical Steps:

1. Open Directory in the Code Editor like VSCode or PyCharm or IntelliJ.
2. Create Two Python Script files and name them server.py and client.py.
3. Paste the server code in the server.py and client code in client.py.
4. Save both files.
5. Open split terminal or just two terminals.

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6. In Terminal 1, Run the server.py file using command `python server.py`.
7. In Terminal 2, Run the client.py file using command `python client.py`.
8. Send a message from client-side using Terminal 2 CLI.
9. Reply back from server-side using Terminal 1 CLI.
10. To end the connection, Send “Over” from the client-side

Code:

Server.py

```
import socket

server_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)

server_socket.bind(("127.0.0.1", 5000))

print("UDP Server started")

print("Waiting for messages...")

while True:

    message, client_address = server_socket.recvfrom(1024)

    message = message.decode()

    print("Client:", message)

    if message == "Over":

        break

    reply = input("Enter reply: ")

    server_socket.sendto(reply.encode(), client_address)

server_socket.close()

print("Server closed")
```

Client.py

```
import socket

client_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
```

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```
client_socket.connect(("127.0.0.1", 5000))
print("UDP Client started")
while True:
    message = input("Enter message: ")
    client_socket.send(message.encode()) ##Encode(): Converts raw string to bytes
    if message == "Over":
        break
    reply = client_socket.recv(1024).decode()
    print("Server:", reply)
client_socket.close()
print("Client closed")
```

Result

```
PS D:\University\Year 3\CN\UDP_Socket_Experiment\UDP_Socket_Experiment> python client.py
UDP Client started
Enter message: Hello, I am Jagrath
Server: You deserve 30/30 marks. Good Child
Enter message: Thank you so much.
Server: Mention not beta, yeh low 30 / 30 in all experiments
Enter message: Over
Client closed
PS D:\University\Year 3\CN\UDP_Socket_Experiment\UDP_Socket_Experiment>
```

```
PS D:\University\Year 3\CN\UDP_Socket_Experiment\UDP_Socket_Experiment> python server.py
UDP Server started
Waiting for messages...
Client: Hello, I am Jagrath
Enter reply: You deserve 30/30 marks. Good Child
Client: Thank you so much.
Enter reply: Mention not beta, yeh low 30 / 30 in all experiments
Client: Over
Server closed
PS D:\University\Year 3\CN\UDP_Socket_Experiment\UDP_Socket_Experiment>
```

Learning Outcome

- Understand the working and characteristics of UDP and its differences from connection-oriented protocols such as TCP.
- Develop and execute a UDP client-server application using Python socket programming for sending and receiving datagrams.